FPU Multi-Block Architecture Design Manual

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1. Overview

1.1 Introduction

This manual describes the design and implementation of a multi-block Floating Point Unit (FPU) with advanced handshake protocols, inspired by the fpnew architecture. The design emphasizes modularity, parallelism, and standard interface compliance.

1.2 Key Features

- **Multi-block Architecture**: Separate blocks for different operation types
- **Standard Handshake Protocol**: AXI4-Stream compatible valid/ready signaling
- **Parallel Execution**: Independent operation blocks enable concurrent processing
- **Modular Design**: Easy to extend with new operations
- **Round-Robin Arbitration**: Fair resource allocation for output conflicts

1.3 Supported Operations

Operation	Code	Description	Block
FLW	3'b000	Floating Load Word	Memory
FSW	3'b001	Floating Store Word	Memory
FADD	3'b010	Floating Add	Arithmetic
FMUL	3'b011	Floating Multiply	Arithmetic
FDIV	3'b100	Floating Divide	Arithmetic

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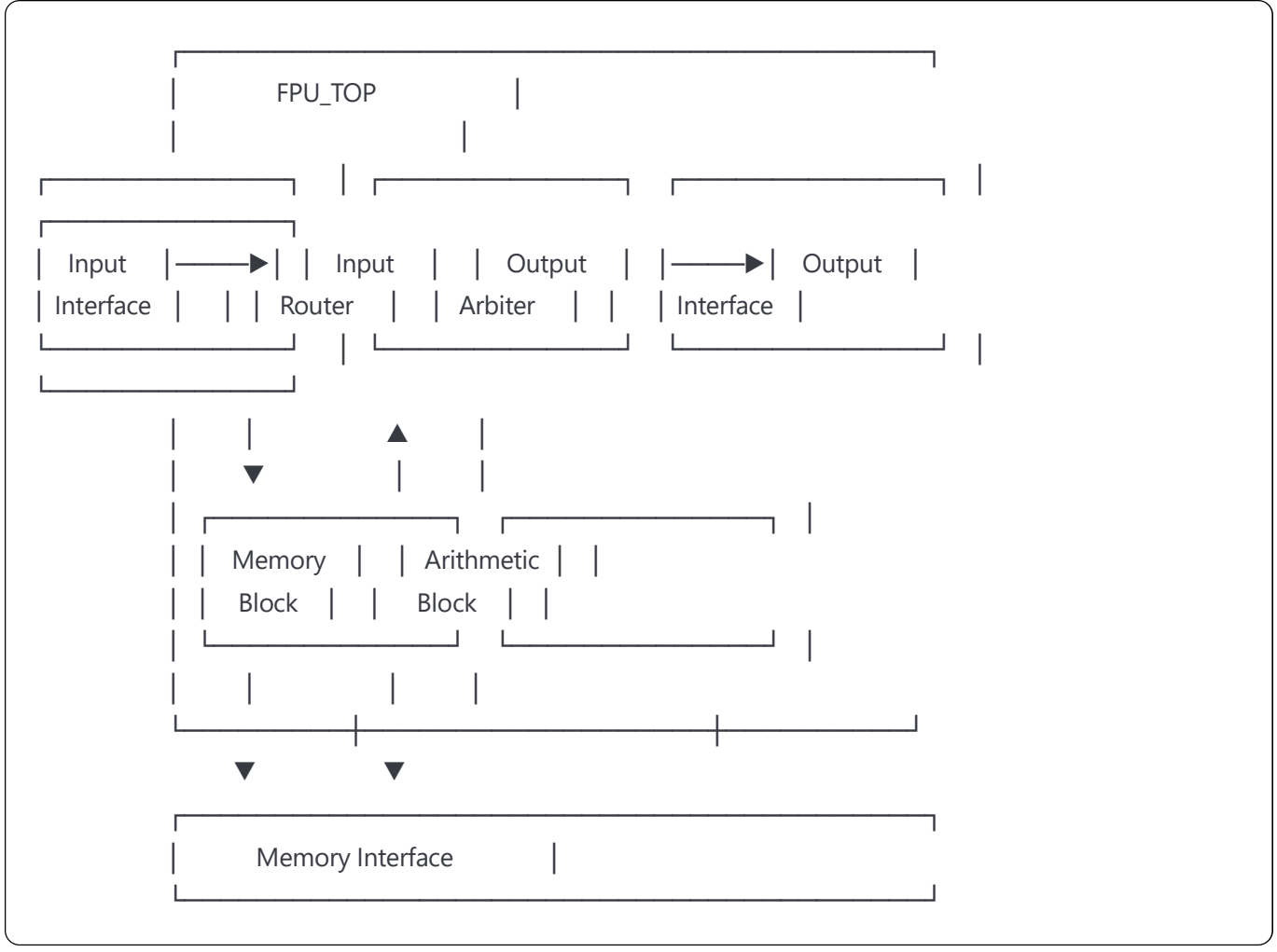
2. Architecture

2.1 Block Diagram



2.2 Operation Group Classification

Memory Operations (OPGRP_MEMORY)

- **FLW**: Load 32-bit floating-point value from memory
- **FSW**: Store 32-bit floating-point value to memory
- **Characteristics**: Memory access required, variable latency

Arithmetic Operations (OPGRP_ARITH)

- **FADD**: Add two floating-point numbers
- **FMUL**: Multiply two floating-point numbers (future)
- **FDIV**: Divide two floating-point numbers (future)
- **Characteristics**: Pure computation, fixed/predictable latency

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3. Interface Specification

3.1 Top-Level Interface

```verilog
module fpu_top (
    // Clock and Reset
    input  logic        clk,          // System clock
    input  logic        rst_n,        // Active-low reset

    // Input Data Interface
    input  operation_e  op_i,         // Operation code
    input  logic [31:0] reg_rs1,      // Source register 1 (integer)
    input  logic [31:0] reg_rs2,      // Source register 2 (integer)
    input  logic [4:0]  rd_addr,      // Destination register address (integer)
    input  logic [31:0] freg_rs1,     // Source register 1 (floating-point)
    input  logic [31:0] freg_rs2,     // Source register 2 (floating-point)
    input  logic [4:0]  frd_addr,     // Destination register address (floating-point)
    input  logic [31:0] imm,          // Immediate value

    // Input Handshake
    input  logic        in_valid_i,   // Input data valid
    output logic        in_ready_o,   // Ready to accept input

    // Memory Interface
    output logic [31:0] mem_addr,     // Memory address
    output logic [31:0] mem_wdata,    // Memory write data
    input  logic [31:0] mem_rdata,    // Memory read data
    output logic        mem_we,       // Memory write enable
    output logic        mem_re,       // Memory read enable
    input  logic        mem_ready,    // Memory ready signal

    // Output Data Interface
    output logic        freg_wb_enable, // FP register write enable
    output logic [4:0]  freg_wb_addr,   // FP register write address
    output logic [31:0] freg_wb_data,   // FP register write data
    output logic        reg_wb_enable,  // Integer register write enable
    output logic [4:0]  reg_wb_addr,    // Integer register write address
    output logic [31:0] reg_wb_data,    // Integer register write data

    // Output Handshake
    output logic        out_valid_o,  // Output data valid
    input  logic        out_ready_i,  // Downstream ready to accept

    // Status
    output logic        busy_o        // FPU busy indicator
);
```

3.2 Signal Descriptions

Input Signals

Signal	Width	Description
clk	1	System clock, rising edge triggered
rst_n	1	Asynchronous reset, active low
op_i	3	Operation code (see operation table)
reg_rs1	32	Integer source register 1 data
reg_rs2	32	Integer source register 2 data
rd_addr	5	Integer destination register address
freg_rs1	32	FP source register 1 data
freg_rs2	32	FP source register 2 data
frd_addr	5	FP destination register address
imm	32	Sign-extended immediate value

Output Signals

Signal	Width	Description
`freg_wb_enable`	1	FP register write enable
`freg_wb_addr`	5	FP register write address
`freg_wb_data`	32	FP register write data
`reg_wb_enable`	1	Integer register write enable
`reg_wb_addr`	5	Integer register write address
`reg_wb_data`	32	Integer register write data
`busy_o`	1	High when FPU is processing

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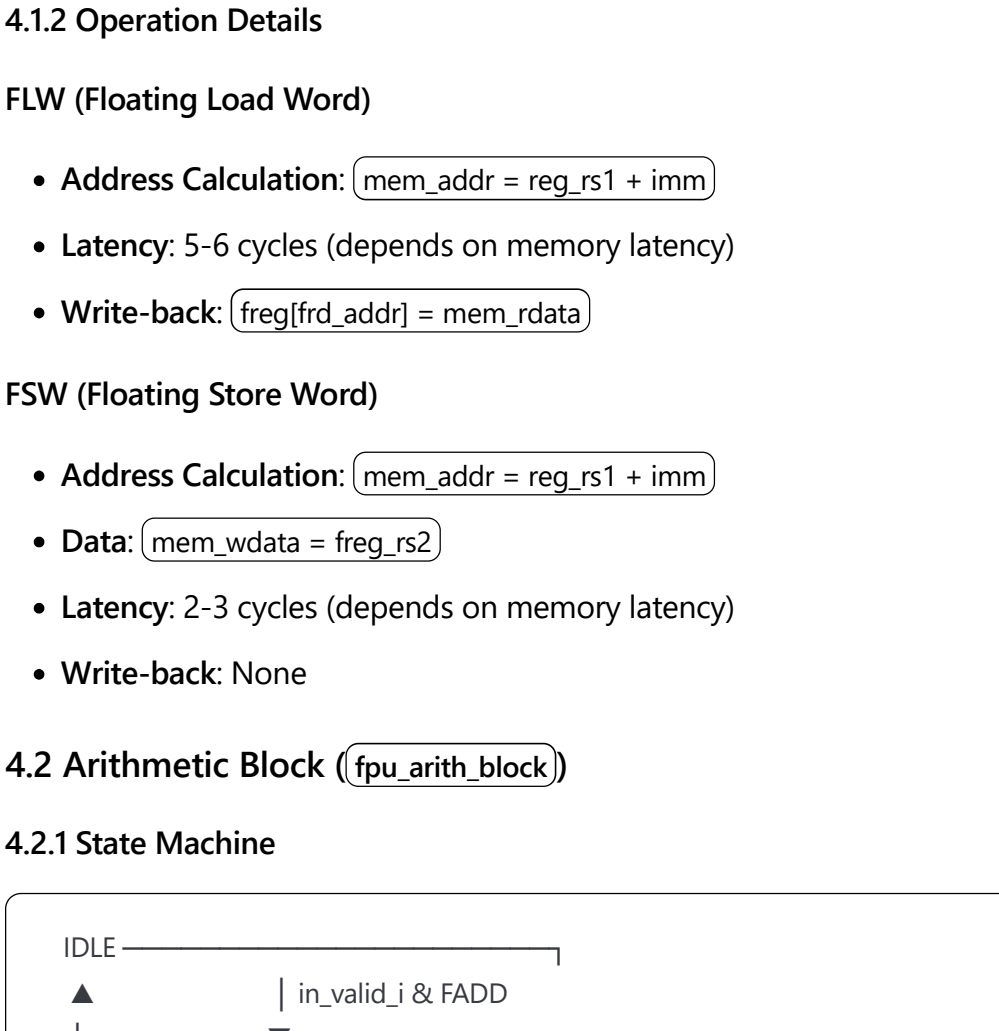
4. Operation Groups

4.1 Memory Block (`fpu_memory_block`)

4.1.1 State Machine



4.1.2 Operation Details

FLW (Floating Load Word)

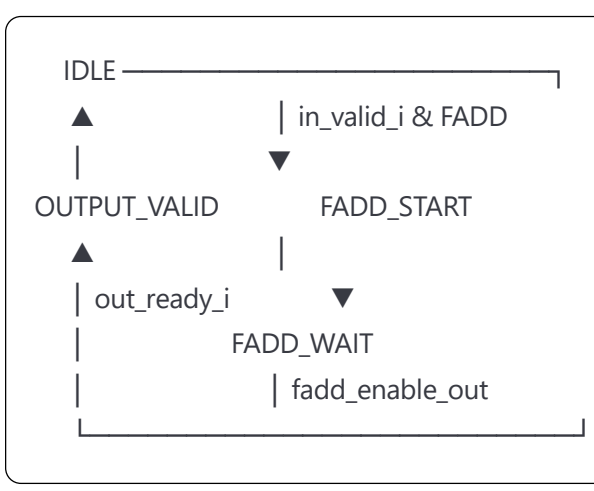
- **Address Calculation**: `mem_addr = reg_rs1 + imm`
- **Latency**: 5-6 cycles (depends on memory latency)
- **Write-back**: `freg[frd_addr] = mem_rdata`

FSW (Floating Store Word)

- **Address Calculation**: `mem_addr = reg_rs1 + imm`
- **Data**: `mem_wdata = freg_rs2`
- **Latency**: 2-3 cycles (depends on memory latency)
- **Write-back**: None

4.2 Arithmetic Block (`fpu_arith_block`)

4.2.1 State Machine



4.2.2 FADD Implementation

- **Inputs**: `freg_rs1`, `freg_rs2`
- **Output**: `fadd_result`
- **Latency**: Depends on fadd_s module (typically 3-5 cycles)
- **Write-back**: `freg[frd_addr] = fadd_result`

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5. Handshake Protocol

5.1 Protocol Overview

The design implements a standard valid/ready handshake protocol similar to AXI4-Stream:

- **Valid**: Source indicates data is available
- **Ready**: Destination indicates ability to accept data
- **Transfer**: Occurs when both valid and ready are high

5.2 Protocol Rules

1. **Valid Independence**: Valid must not depend on ready
2. **Ready Dependency**: Ready may depend on valid
3. **Stability**: Valid data must remain stable until transfer
4. **No Combinational Paths**: Avoid ready→valid combinational loops

5.3 Input Protocol

```verilog
// Transfer occurs when both signals are high
wire input_transfer = in_valid_i & in_ready_o;

// Ready generation (independent of valid)
assign in_ready_o = (state == IDLE);

// Valid handling
if (input_transfer) begin
    // Accept new operation
    // Transition to processing state
end
```

5.4 Output Protocol

```verilog
// Transfer occurs when both signals are high
wire output_transfer = out_valid_o & out_ready_i;

// Valid generation
assign out_valid_o = (state == OUTPUT_VALID);

// Ready handling
if (output_transfer) begin
    // Complete current operation
    // Transition to idle state
end
```

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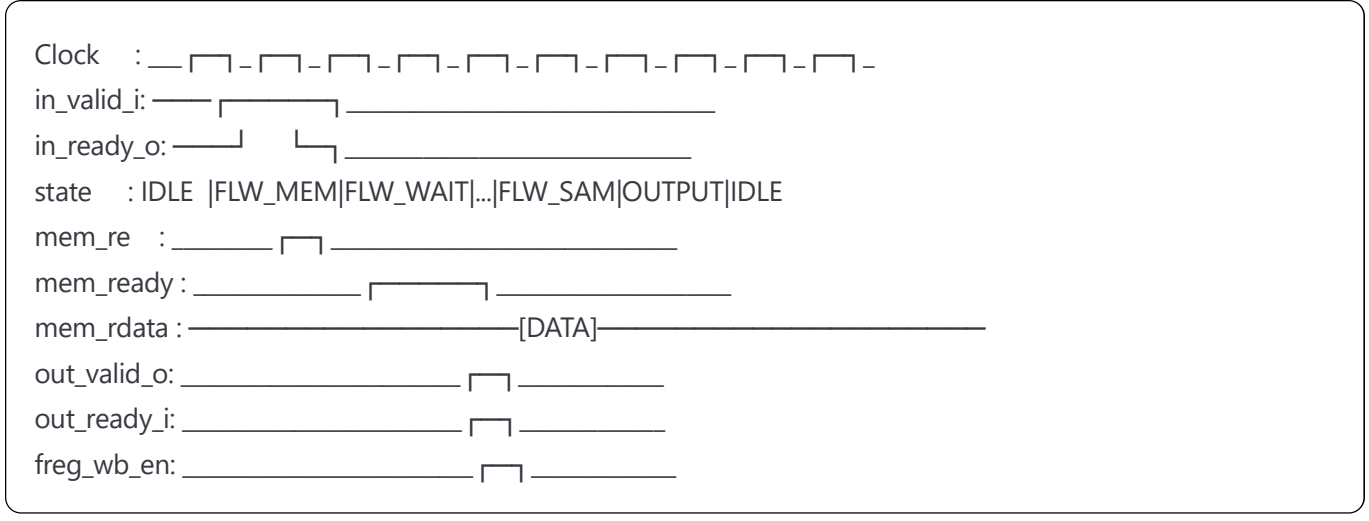
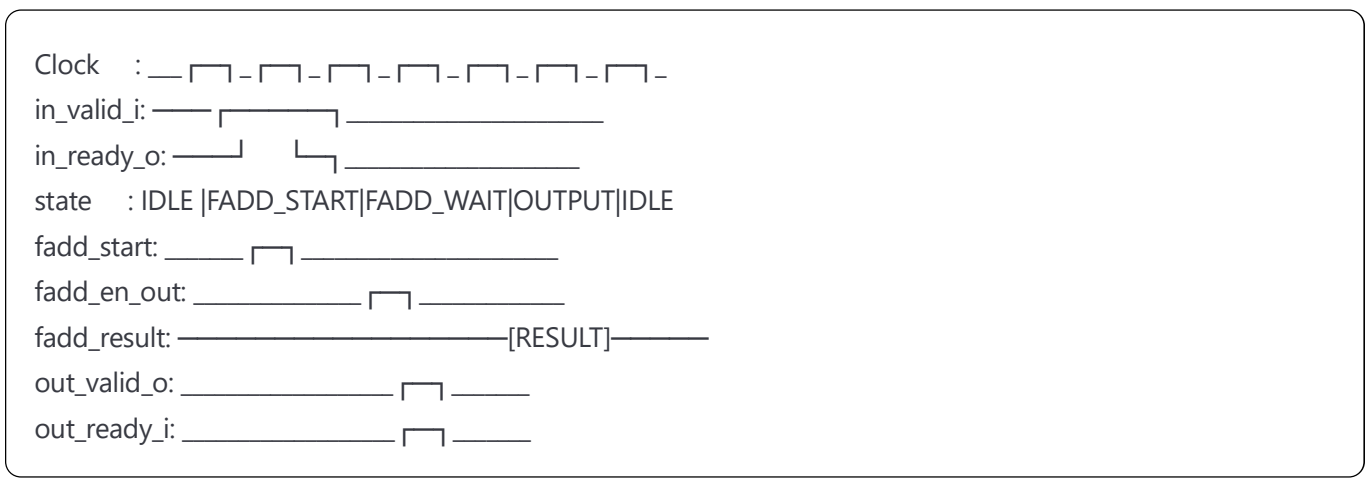
6. Timing Diagrams

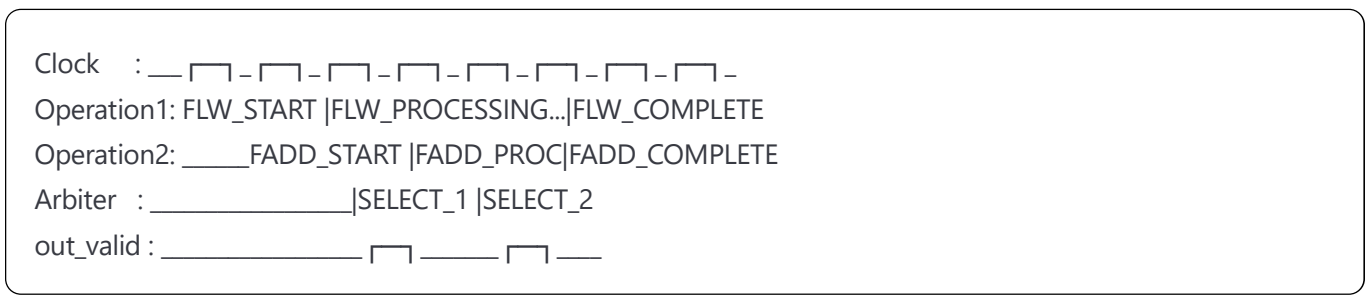
6.1 FLW Operation Timing



6.2 FADD Operation Timing



6.3 Parallel Operations



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7. Design Implementation

7.1 Key Design Patterns

7.1.1 State Machine Template

verilog

```verilog
// State definition
typedef enum logic [2:0] {
    IDLE    = 3'd0,
    STAGE1  = 3'd1,
    STAGE2  = 3'd2,
    OUTPUT  = 3'd3
} state_e;

state_e state, next_state;

// Next state logic (combinational)
always_comb begin
    next_state = state;
    case (state)
        IDLE: if (condition) next_state = STAGE1;
        STAGE1: next_state = STAGE2;
        STAGE2: next_state = OUTPUT;
        OUTPUT: if (out_ready_i) next_state = IDLE;
        default: next_state = IDLE;
    endcase
end

// State register
always_ff @(posedge clk or negedge rst_n) begin
    if (!rst_n) state <= IDLE;
    else state <= next_state;
end

// Output logic
always_ff @(posedge clk or negedge rst_n) begin
    if (!rst_n) begin
        // Reset outputs
    end else begin
        // Default values
        case (state)
            STAGE1: // Stage 1 actions
            STAGE2: // Stage 2 actions
            OUTPUT: out_valid_o <= 1'b1;
        endcase
    end
end
```

7.1.2 Handshake Interface Pattern

verilog

```verilog
// Input interface
assign in_ready_o = (state == IDLE);
wire input_fire = in_valid_i & in_ready_o;

// Output interface
assign out_valid_o = (state == OUTPUT);
wire output_fire = out_valid_o & out_ready_i;

// Usage in state machine
if (input_fire) begin
    // Accept new request
end

if (output_fire) begin
    // Complete current request
end
```

7.2 Arbitration Logic

7.2.1 Round-Robin Algorithm

```verilog
// Round-robin pointer
logic [$clog2(NUM_INPUTS)-1:0] rr_pointer;

always_ff @(posedge clk or negedge rst_n) begin
    if (!rst_n) begin
        rr_pointer <= '0;
    end else if (output_fire) begin
        rr_pointer <= (grant_idx + 1) % NUM_INPUTS;
    end
end

// Mask generation
logic [NUM_INPUTS-1:0] mask;
always_comb begin
    mask = '0;
    for (int i = 0; i < NUM_INPUTS; i++) begin
        if (i >= rr_pointer) mask[i] = 1'b1;
    end
end

// Priority encoding with mask
logic [NUM_INPUTS-1:0] masked_req = req_i & mask;
// If masked_req == 0, use req_i directly
```

7.3 Common Pitfalls and Solutions

7.3.1 Combinational Loops

Problem: Ready signal depends on valid signal

```verilog
// WRONG
assign in_ready_o = in_valid_i & (state == IDLE);
```

Solution: Make ready independent of valid

```verilog
// CORRECT
assign in_ready_o = (state == IDLE);
```

7.3.2 Data Stability

Problem: Data changes before handshake completes

```verilog
// WRONG - data may change
always_ff @(posedge clk) begin
    output_data <= new_data;  // Changes every clock
    out_valid_o <= valid_condition;
end
```

Solution: Hold data stable until transfer

```verilog
// CORRECT - data held stable
always_ff @(posedge clk) begin
    if (state == COMPUTE) begin
        output_data <= new_data;
        out_valid_o <= 1'b1;
    end else if (output_fire) begin
        out_valid_o <= 1'b0;
        // Keep output_data stable
    end
end
```

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8. Verification Guide

8.1 Testbench Architecture

```verilog
module fpu_tb;
  // DUT instantiation
  fpu_top dut (...);

  // Clock generation
  always #5 clk = ~clk;

  // Test scenarios
  initial begin
    reset_sequence();
    test_flw_operation();
    test_fsw_operation();
    test_fadd_operation();
    test_parallel_operations();
    test_backpressure();
    $finish;
  end

  // Monitor and checker
  property handshake_rule;
    @(posedge clk)
    $rose(out_valid_o) |-> ##[0:$] out_ready_i;
  endproperty

  assert property (handshake_rule);
endmodule
```

8.2 Test Cases

8.2.1 Basic Functionality Tests

1. **Reset Test**: Verify all outputs are in reset state
2. **Single Operation Tests**: Test each operation individually
3. **Data Path Tests**: Verify correct data flow
4. **Edge Cases**: Test boundary conditions

8.2.2 Protocol Compliance Tests

1. **Handshake Protocol**: Verify valid/ready behavior
2. **Back-pressure**: Test with slow downstream
3. **Burst Transfers**: Multiple consecutive operations
4. **Random Delays**: Insert random ready delays

8.2.3 Performance Tests

1. **Throughput**: Measure operations per cycle
2. **Latency**: Measure end-to-end delay
3. **Parallel Execution**: Verify concurrent operations
4. **Arbitration Fairness**: Check round-robin behavior

8.3 Assertion Examples

```systemverilog
// Protocol assertions
property valid_stable;
    @(posedge clk)
    $rose(out_valid_o) |->
    (out_valid_o && $stable(freg_wb_data)) until out_ready_i;
endproperty

property no_combinational_ready;
    @(posedge clk)
    !$rose(in_valid_i) |-> $stable(in_ready_o);
endproperty

// Functional assertions
property flw_correct_addr;
    @(posedge clk)
    (in_valid_i && in_ready_o && op_i == OP_FLW) |->
    ##[1:10] (mem_re && mem_addr == $past(reg_rs1 + imm, 1));
endproperty
```

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9. Performance Analysis

9.1 Latency Analysis

Operation	Best Case	Worst Case	Average
FLW	5 cycles	7 cycles	6 cycles
FSW	2 cycles	4 cycles	3 cycles
FADD	3 cycles	5 cycles	4 cycles

9.2 Throughput Analysis

9.2.1 Single Block Utilization

- **Memory Block**: Can start new operation every 6 cycles (average)
- **Arithmetic Block**: Can start new operation every 4 cycles (average)
- **Overall**: Limited by slowest operation

9.2.2 Parallel Execution Benefits

- **Without Parallelism**: Sequential execution of different operation types
- **With Parallelism**: Memory and arithmetic operations can overlap
- **Improvement**: Up to 40% throughput increase for mixed workloads

9.3 Resource Utilization

```
Resource Type    | Original FPU | Multi-block FPU | Overhead
-----------------|--------------|-----------------|----------
Logic Cells      | 1,200        | 1,450           | +20.8%
Registers        | 800          | 950             | +18.8%
Memory Bits      | 0            | 0               | 0%
DSP Blocks       | 2            | 2               | 0%
```

9.4 Power Analysis

- **Dynamic Power**: Increased due to additional logic
- **Static Power**: Minimal increase
- **Power Efficiency**: Better performance per watt due to higher throughput

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10. Appendices

Appendix A: Signal Reference

A.1 Complete Signal List

[Comprehensive table of all signals with descriptions, widths, and timing requirements]

A.2 Register Map

[If applicable, memory-mapped control/status registers]

Appendix B: Protocol Details

B.1 AXI4-Stream Compliance

[Detailed comparison with AXI4-Stream protocol]

B.2 Custom Extensions

[Any protocol extensions beyond standard]

Appendix C: Design Trade-offs

C.1 Area vs Performance

- **Higher Area**: Multiple state machines, arbitration logic
- **Higher Performance**: Parallel execution, better throughput
- **Decision**: Performance prioritized for multi-operation workloads

C.2 Latency vs Throughput

- **Individual Latency**: Slightly increased due to handshake overhead
- **System Throughput**: Significantly improved due to parallelism
- **Decision**: Optimized for throughput-sensitive applications

Appendix D: Future Enhancements

D.1 Planned Features

1. **FMUL Implementation**: Add floating-point multiplication
2. **FDIV Implementation**: Add floating-point division
3. **Pipeline Optimization**: Reduce individual operation latency
4. **SIMD Support**: Vector floating-point operations

D.2 Architecture Extensions

1. **Multi-port Memory**: Parallel memory operations
2. **Register Scoreboarding**: Advanced dependency tracking
3. **Branch Prediction**: For conditional FP operations

Appendix E: Reference Materials

E.1 Related Standards

- IEEE 754: Floating-Point Arithmetic Standard
- AXI4-Stream Protocol Specification
- RISC-V F Extension Specification

E.2 Bibliography

- "Computer Architecture: A Quantitative Approach" - Hennessy & Patterson
- "Digital Design and Computer Architecture" - Harris & Harris
- "FPGA Prototyping by SystemVerilog Examples" - Pong P. Chu

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